

03

BIOLOGY

After analysing the previous year question papers, we have seen that 40-45 questions are asked from science section, out of these 10-15 questions were asked from Biology. From biology section, around 2-3 questions are asked from classification, genetics and evolution, 3-4 questions from human physiology, and 1 to 2 questions each from topics like cell, human health and diseases and applied biology.



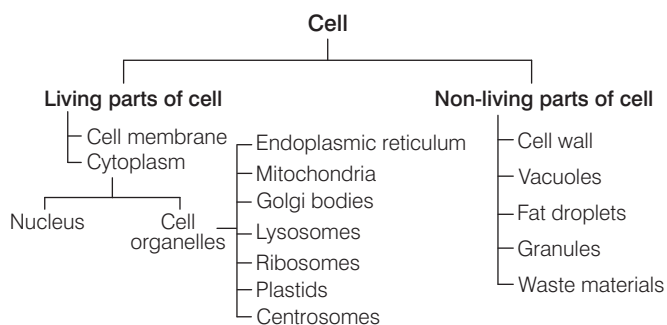
CELL : THE UNIT OF LIFE

Biology is the branch of science concerned with the study of living organisms. The term 'biology' is coined by **Lamarck** and **Treviranus**. The study of plants is called **Botany**, while study of animals is called **Zoology**.

THE CELL

Cell is the basic structural and functional unit of all the living organisms. The word 'cell' comes from the Latin word *cellula* which means a 'small room'.

- Living organisms may consist of one or more cells, according to this there are two types of organisms, i.e. **unicellular** (composed of single cell) and **multicellular** (composed of many cells).



- Robert Hooke** in 1665 discovered the cell. He is the Father of Cytology.
- Cell theory was given by **Schleiden** and **Schwann**.
- Matthias Jacob Schleiden** (1838) a German botanist said that, all plants are composed of different kinds of cells, which in turn form plant tissues.
- Theodor Schwann** (1839) said that, all animals and plants are made up of cells. Cell theory is not applicable for viruses.
- According to **Virchow** 'new cells arise from pre-existing cells', i.e. *Omnis cellula-e-cellula*.
- Smallest cell is mycoplasma or Pleuropneumonia Like Organism (PPLo). Largest cell is ostrich's egg and the longest cell is nerve cell.
- The smallest human cell is the red blood cell. Bacteria was discovered by **AV Leeuwenhoek**.
- First compound microscope was invented by **Jansen** and **Jansen**.
- Electron microscope was invented by **Knoll** and **Ruska**.
- Vital stains are used for staining living cells or living parts of cell, e.g. Janus green for mitochondria.
- Protoplasm is the living component of cell that includes cytoplasm, cell organelles and nucleoplasm.
- Cytoplasm term was given by **Strasburger**.

- Mitochondria, chloroplast, nucleus are double-membraned structures.
- Lysosome, Golgi body, Endoplasmic Reticulum (ER) are single-membraned structures.
- Centrosome, flagella, ribosome are non-membranous in structure.
- Glycogen is referred to as animal starch.

Some Discoveries about the Cell

Scientist	Discovery
Robert Brown (1831)	Nucleus
Purkinje (1837)	Named protoplasm to be a content of cell
W Flemming (1882)	Observed mitotic cell division, splitting of chromosome and named the term chromatin
F Miescher	Discovered the nucleic acid
C Benda (1897)	Mitochondria
Beneden and Boveri (1887)	The chromosome number is constant in a species
Camillo Golgi (1891)	Golgi body
WS Sutton (1902)	Significance of reductional division
F Meves (1904)	Mitochondria in plant cell
JB Farmer and Moore (1905)	Named meiosis
FA Johanssen (1909)	Chiasma formation
TH Morgan (1933)	Role of chromosomes in heredity

Types of Cells

On the basis of the organisation, complexity and variety, all cells can be grouped into two types:

- (i) **Prokaryotic cell** It is a simpler cell, lacking a true nucleus, i.e. nuclear membrane is absent in these cells, e.g. bacteria, mycoplasma, etc.



Bacteria are the microscopic prokaryotic organisms whose single cell have neither a membrane enclosed nucleus nor other membrane enclosed organelles like mitochondria and chloroplasts.

- The bacterial cell wall is strong and rigid.
- Murein or peptidoglycan or mucopeptide is present in the cell wall.

Bacteria are mainly of two types:

- (i) **Archaeobacteria** They are obligate anaerobes.
 (ii) **Eubacteria** They are true bacteria.

Some examples are as follow:

- *Staphylococcus aureus* causes boils, pneumonia, food poisoning and other diseases.
- *Staphylococcus thermophilus* gives yoghurt its creamy flavour.
- *S. lactis* often added to milk for the production of cheese.

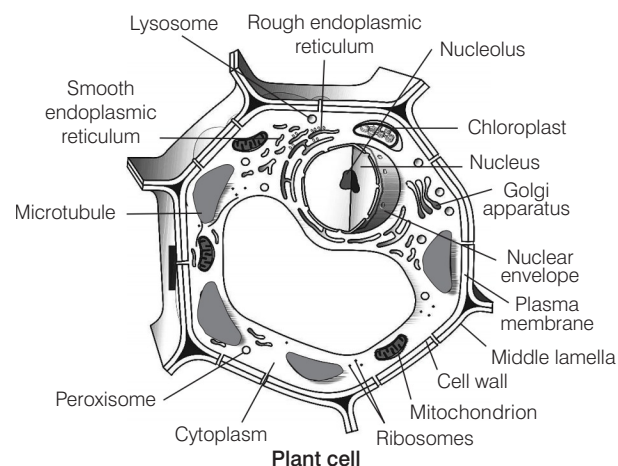
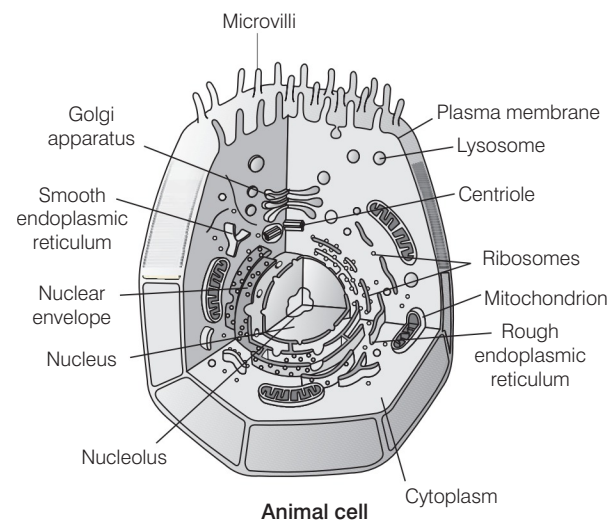
- (ii) **Eukaryotic cell** It is a complex cell with double membrane bound nucleus, e.g. plant and animal cells. Eukaryotic organisation is seen in all the protists, plants, fungi and animals.

Differences between Prokaryotic and Eukaryotic Cell

Prokaryotic cell	Eukaryotic cell
Simplest and primitive in nature.	Advanced and comparatively complex in nature.
Nucleus is absent.	Nucleus is present.
Membrane bound cell organelles are absent.	Membrane bound cell organelles are present.
Single naked chromosome is present.	Many chromosomes are present.
Cell division is direct.	Cell division by mitosis or meiosis.

Structure of Animal and Plant Cell

Animal and plant cell structures are described below



Differences between Plant and Animal Cell

<i>Plant cell</i>	<i>Animal cell</i>
Cell wall is present in plant cell.	Cell wall is usually absent.
Plastids are found.	Plastids are usually absent.
Centrioles are found in some algae and fungi.	Centrioles are found in all cells.
Big vacuoles are present but they are less in number.	Vacuole is very small in size but present in more number.
Cell shape is usually round or irregular.	Cell shape is rectangular and fixed.

Composition of Cell

Representation of the chemical composition of living tissue is as follows:

<i>Component</i>	<i>The total cellular mass in %</i>	<i>Component</i>	<i>The total cellular mass in %</i>
Water	70-90%	Lipids	2%
Proteins	10-15%	Nucleic acids	5-7%
Carbohydrates	3%	Ions	1%

Cell Organelles

There are some organelles as follow:

Cell Wall

It is a rigid and solid covering that gives shape and strong structural support to the cell. It is a non-living structure, which forms the outer covering of the plasma membrane in plants and fungi, but is absent in animal cells. Cell wall consists of following parts:

- **Middle lamella** It is composed of calcium pectate (chief component) and magnesium pectate.
- **Primary cell wall** It is composed of cellulose, hemicellulose and lignin. Rigidity of cell wall is due to lignin.
- **Secondary cell wall** It is composed of cellulose and hemicellulose.
- **Tertiary cell wall** It is present in tracheids of gymnosperms.
- Bacterial cell wall is composed of **peptidoglycan** or **murein**. Bacterial cell wall contains D-amino acids.
- Fungal cell wall is composed of chitin.

Functions of Cell Wall

- It helps in preventing cell from bursting or collapsing and allows the material to pass in and out of the cell.
- It wards off the attack of pathogens like viruses, bacteria, fungi and protozoans.

Cell Membrane or Plasmalemma

It is the innermost layer of cell envelope that is semipermeable in nature and is responsible for the interaction of cell with the outside environment.

- It consists of protein (52%), lipid (40%) and carbohydrate (5-10%). It is elastic in nature because of the presence of lipids.
- The most popular theory 'fluid mosaic model' of membrane structure was given by **Singer** and **Nicolson**.
- It states that integral and peripheral proteins are found in cell membrane.
- **Pinocytosis** (cell drinking) In this process, cells ingest extracellular fluid.
- **Phagocytosis** (cell eating) Engulfing and digestion of extracellular solid food by cell.

Nucleus

The term 'Nucleus' is given by **Waldeyer** (1888). Nucleus is a specialised largest principal cell organelle of the cell, which contains all genetic information for controlling all essential processes related to metabolism and transmission. It was first discovered by **Robert Brown** in 1831.

- Most of the eukaryotic cells are uninucleate except *Paramecium*, which is binucleate.
- Nucleus is made up of nuclear membrane, nucleoplasm, nucleolus, chromatin. Chromatin is composed of nucleic acid and protein (histone and non-histone chromosome).
- Chromosome of eukaryotic cell is composed of DNA + histone protein + some amount of RNA. Normally each cell contains 23 pair of chromosomes in humans.
- Nucleic acid, the genetic material was discovered by **Friedrich Miescher**. The term 'Nucleic acid' was given by **Altman**.

Chromosome

During the cell division, chromatin shrinks (compressed) and gets divided into various smaller, thick and consolidated form, known as chromosomes.

Chromosome Number in Some Organisms

<i>Organisms</i>	<i>Chromosome number in each body cell</i>	<i>Organisms</i>	<i>Chromosome number in each body cell</i>
Roundworm	2	Mouse	40
Mosquito	6	Rat	42
<i>Drosophila</i>	8	Human beings	46
Garden pea	14	Potato	48
Onion	16	Dog	64
Maize	20	Pigeon	80
Frog	24	Gold fish	100
Rice	26	<i>Amoeba</i>	250
Sunflower	34		

Study of Cell Organelles

Cell organelle	Nature/Composition	Function	Present (P)/Absent (A)	
			Animal cell	Plant cell
Cell membrane	Double layer of lipid molecules with proteins molecules, differentially permeable.	Gives shape to the cell, maintains distinct composition of the cell.	P	P
Cell wall	Freely permeable, made up of fibrous polysaccharide (cellulose). Bacteria and unicellular eukaryotes have peptidoglycan compound in their cell wall.	Gives strength and rigidity to the cell.	A	P
Cytoplasm	Consists of aqueous ground substance, having starch, glycogen, etc.	Provides medium for suspension of materials and cell organelles.	P	P
Vacuoles	Fluid-filled spaces without any definite shape and size. Larger in plant cell.	Store water, minerals, salts, food pigments, wastes, etc.	P	P
Nucleus	Oval-shaped, bounded by double membrane having pores possess genetic material.	Controls the metabolic activities, regulates cell division, synthesises and stores proteins, also called as the controller of the cells .	P	P
Mitochondria	Bounded by a double membrane, found in cytoplasm. They contain DNA also.	Cellular respiration or oxidation of food, ATP is stored in it, so called as powerhouse of the cell .	P	P
Endoplasmic Reticulum (ER)	Membranous network of fluid-filled with lumen with or without ribosomes, absent in RBCs.	Forms supporting skeletal framework of the cell also called skeleton of the cell .	P	P
Ribosomes	Dense and spherical particles, found either attached to endoplasmic reticulum or freely in cytoplasm. It is the smallest non-membranous organelles.	Play a vital role in protein synthesis also called protein factory of the cell .	P	P
Lysosomes	Tiny spherical bag-like structures found in cytoplasm with hydrolytic enzymes.	Helps in intracellular digestion, digest worn out organelles, also called suicidal bags of the cell.	P	P
Chloroplasts	Sac-like double walled with protein matrix generally called plastids , contain photosynthetic pigments called chlorophyll and carotenoids . These also contains DNA.	Sites of photosynthesis, give colour to organ, also called as kitchen of the cell .	A	P
Golgi bodies	Consist of tubules and vesicles.	Secrete hormones, enzymes, involved in synthesis of cell wall, plasma membrane, etc.	P	P
Centriole	Hollow and cylindrical structures made up of microtubules.	Helps in cell division in animal cell.	P	A
Microtubules	Tubulin protein	Spindle formation	P	P

CELL CYCLE

- Cell cycle is an orderly sequence of events or stages by which a cell duplicates its genome, synthesise the other constituents of the cell and eventually divides into two daughter cells. It is a cyclic process, which involves cell growth and cell division.
- Cell division** is an essential process in all living organisms. The mode of cell division is fundamentally similar in all organisms. During cell division, the processes like DNA replication and cell growth must take place in a sequential coordinated manner to ensure the correct division and formation of progeny cells with intact genome. Cell cycle involves two phases, i.e. **interphase** followed by **M-phase**.

1. Interphase

It is the period between the end of one cell division to the beginning of the next cell division (i.e. between true successive M-phase). Interphase is further divided into following three substages on the basis of various synthetic activity.

- G₁-phase (Gap 1)** It corresponds to the duration between the mitosis (M-phase) and initiation of replication of DNA. Cell grows, becomes metabolically active, but does not undergo DNA replication. Some time after G₁-phase cell may enter S-phase and continue the cell cycle or it may enter into G₀-phase. In G₀-phase the cells leave cell cycle.
- S-phase (Synthesis)** It is known to be the phase, in which actual synthesis or replication of DNA takes place. In case of animal cell, during S-phase DNA replication begins inside the nucleus, while the duplication of centrioles takes place in the cytoplasm.
- G₂-phase (Gap 2)** This phase is also called **post-synthetic** or **pre-mitotic phase**. During this stage, the synthesis of DNA stops and proteins required for M-phase are being synthesised, while the growth of the cells continues. It prepares the cell to undergo divisions.

2. M-phase

M-phase is the actual phase of cell division. It may occur in two way depending upon the type of cell, it is taking place in, i.e. mitosis (vegetative cell) or meiosis (generative cell).

(A) Mitosis

A type of cell division, in which chromosomes replicate themselves and get equally distributed into daughter nuclei, i.e. the chromosome number in the parental and the progeny becomes same.

The four phases of mitosis are:

- (i) **Prophase** Chromosome deviation in chromatids starts and nuclear membrane and nucleolus disappear.
- (ii) **Metaphase** Chromosomes line up in the middle and spindle fibres from centriole connect to each chromatid (half of chromosome).
- (iii) **Anaphase** Chromatids are pulled apart to separate poles and the membrane begins to pinch off in the middle.
- (iv) **Telophase** Complete division of cytoplasm (cytokinesis) and two cells are formed. Two daughter cells, which are formed by mitosis have same/equal number of chromosomes. It results in formation of somatic cells.

(B) Meiosis

A type of cell division that results in the formation of four daughter cells each with half the chromosome number of the parent cell.

In the process of meiosis, there are two main stages:

Meiosis-I

It is further divided into four phases:

- (i) **Prophase-I** The longest phase of meiosis-I. It is composed of five substages:
 - (a) **Leptotene** Chromatins condense to form chromosome.
 - (b) **Zygotene** Pairing of homologous chromosomes called **synapsis**, in this phase, synaptonemal complex is formed.
 - (c) **Pachytene** Crossing over between non-sister chromatids.
 - (d) **Diplotene** Longest duration phase, in which chiasmata formation takes place.
 - (e) **Diakinesis** It involves slipping of chiasmata called **terminalisation**.

- (ii) **Metaphase-I** Lining of chromosomes in the middle. Formation of spindle apparatus takes place and each centromere is joined by chromosomal fibres.
- (iii) **Anaphase-I** Separation of homologous chromosomes (disjunction) to form two cells with haploid number of the chromosomes.
- (iv) **Telophase-I** This is the last stage of meiosis-I, two cells with n number of chromosomes are formed.

Meiosis-II

It is very similar to mitosis.

- (i) **Prophase-II** The shortening and thickening of the chromatids.
- (ii) **Metaphase-II** Lining of chromosomes in the middle.
- (iii) **Anaphase-II** Separation of chromosomes into chromatids.
- (iv) **Telophase-II** Uncoiling and lengthening of the chromosomes, four daughter cells with haploid set of chromosomes are formed.

Differences between Mitosis and Meiosis

Mitosis	Meiosis
Found in somatic cells.	Found in diploid reproductive cells.
Forms two identical diploid cells.	Forms four haploid cells.
Every chromosome behaves independently.	Homologous chromosomes show pairing.
Chromosome number remains constant.	Chromosome number becomes half.
Crossing over does not occur.	Crossing over occurs.



IMPORTANT POINTS

- Chromoplasts are yellow, orange and red coloured.
- Leucoplasts are colourless plastids.
- **Golgi bodies** are absent in bacteria, blue-green algae, mature sperm and RBCs.
- Dinomitosis is failure of meiosis-II to occur after meiosis-I. It solely occurs in the class-Dinophyceae, a class of Alga with chlorophyll-*c*.

CLASSIFICATION OF PLANTS AND ANIMALS

TAXONOMY

The classification results in an organised system, which is essential for the naming and cataloging of future specimens and reflects inter-relationships between living organisms. Biologists use method of classification to understand the evolutionary history and it provides an organised way of studying characteristics of living organisms.

- **Classification** means the ordering of organisms into groups. The branch of science that deals with the principles and procedures of biological classification is called **taxonomy**.
- **Carolus Linnaeus** is the Father of Taxonomy. '*Systema Naturae*' and '*Species Plantarum*' books are written by Carolus Linnaeus.